

EXCEL DATA ANALYSIS PROJECT

Classic Models Inc.

Sales Performance Dashboard

A complete end-to-end analytics project for Excel learners:
clean 7 raw data files, connect them with lookups, analyse with
pivot tables, and present findings in a one-page interactive dashboard.

TOOL

Microsoft Excel only

LEVEL

Beginner - Intermediate

DURATION

10-14 days

DELIVERABLE

1 workbook + dashboard

Dataset: 7 CSV files | 122 customers | 326 orders | 110 products | 2003 - 2005

Prepared by your mentor | Includes step-by-step guide, dashboard blueprint, challenges & grading rubric

1. Project Brief

You have just joined **Classic Models Inc.**, a company that sells scale models of classic cars, motorcycles, planes, ships and trains to retailers around the world. Management has handed you the company's raw data — 7 CSV files exported from their order system — and asked one question: **“How is our business really doing, and where should we focus next year?”**

Your job is to turn these raw files into a single, polished Excel workbook that ends in a **one-page interactive dashboard** a manager could use without touching a single formula.

What you will practise

- **Data cleaning:** handling NULL values, fixing data types, converting text dates to real dates.
- **Connecting tables:** XLOOKUP / VLOOKUP and INDEX-MATCH to combine 7 files into one analysis sheet.
- **Calculated columns:** Revenue, Profit, Year, Month, shipping delay and more.
- **Aggregation:** Pivot Tables, SUMIFS / COUNTIFS / AVERAGEIFS.
- **Visualisation:** column, bar, line and doughnut charts; sorting and formatting them properly.
- **Interactivity:** Slicers and a Timeline connected to every pivot.
- **Business storytelling:** writing insights, not just numbers.

Final deliverables (what you submit)

#	Deliverable	Details
1	One Excel workbook (.xlsx)	All 7 data sheets + Analysis sheet + Pivots sheet + Dashboard sheet
2	Dashboard (one screen)	5 KPI cards, 5-6 charts, 4-5 slicers, consistent colour theme
3	Insights sheet	8-10 written findings + 3 recommendations for management
4	5-minute walkthrough	Present your dashboard and defend your numbers (viva)

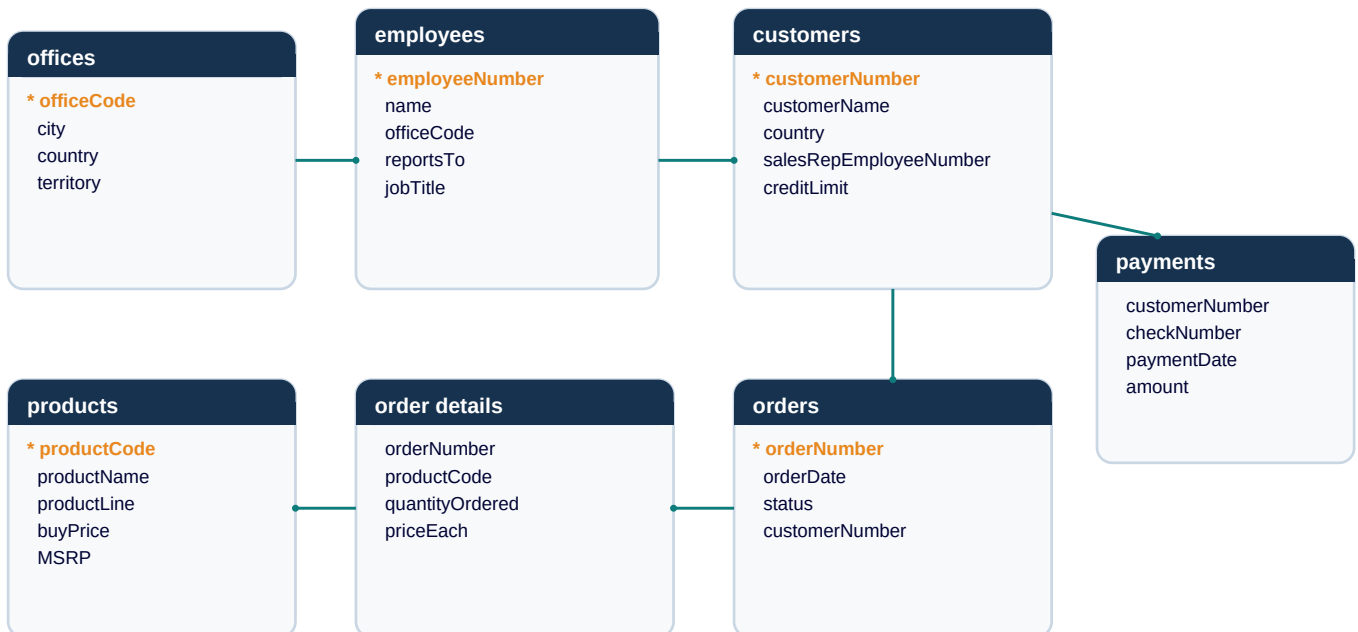
Rule for this project: Excel only. No SQL, no Power BI, no Python. Everything here is achievable with formulas, Pivot Tables and charts. (Excel 2016 or later recommended; use VLOOKUP if your version has no XLOOKUP.)

2. Know Your Data

Before writing a single formula, spend 30 minutes just reading the files. The 7 CSVs are pieces of one connected business database:

File	Rows	What it contains	Key column(s)
customers.csv	122	Every customer: name, city, country, credit limit, assigned sales rep	customerNumber
employees.csv	24	Staff, their job titles, who they report to, office they sit in	employeeNumber
offices.csv	7	Company offices and their territory (NA, EMEA, APAC, Japan)	officeCode
orders.csv	326	One row per order: dates, status, which customer placed it	orderNumber
order details.csv	2,996	One row per product per order: quantity and the price actually charged	orderNumber + productCode
products.csv	110	Product catalogue: product line, cost (buyPrice), list price (MSRP), stock	productCode
payments.csv	273	Cheques received from customers with date and amount	customerNumber + checkNumber

How the tables connect (relationship map)



* = key column used to connect tables (lookup key). Lines show which tables join to which.

Read the map like this: an **order** belongs to a **customer**; each order has several **order detail** lines; each line points to a **product**; each customer has a sales rep (**employee**), who sits in an **office**; customers send **payments**. Whenever you need a column from another table, you will look it up using the key column marked with *.

Important business fact: revenue does NOT live in orders.csv. Revenue = quantityOrdered × priceEach, calculated line by line in *order details*. This is the single most common mistake in this project.

3. Phase 1 — Set Up the Workbook (Day 1)

Goal: one workbook, seven clean sheets, everything formatted as Excel Tables.

1. Create a new blank workbook. Save it immediately as **ClassicModels_Analysis_YourName.xlsx**.
2. Import each CSV: **Data** → **Get Data** → **From Text/CSV** (or simply open the CSV and copy-paste into a new sheet). One sheet per file.
3. Rename the sheets: **Customers, Employees, Offices, Orders, OrderDetails, Products, Payments**.
4. Convert each range into an Excel Table: click inside the data, press **Ctrl+T**, tick "My table has headers". Then on the Table Design tab give each table a proper name: **tblCustomers, tblOrders, tblOrderDetails, tblProducts, tblEmployees, tblOffices, tblPayments**.
5. Add a **ReadMe** sheet at the front listing your name, date, and a one-line description of every sheet.

Why Tables? Formulas written against Tables auto-fill the whole column, ranges auto-expand, and Pivot Tables stay connected when data changes. Professionals always work this way.

4. Phase 2 — Clean the Data (Days 2-3)

Raw exports are never perfect. Fix these specific issues and note each fix in your ReadMe sheet:

Issue	Where you'll find it	How to fix it in Excel
Text 'NULL' instead of blanks	customers (addressLine2, state), orders (comments, shippedDate), employees (reportsTo)	Select column → Ctrl+H → Find 'NULL' → Replace with nothing. Do this column by column, not on whole sheet, to stay safe.
Dates stored as text	orders (orderDate, requiredDate, shippedDate), payments (paymentDate)	Check with =ISNUMBER(A2). If FALSE: select column → Data → Text to Columns → Finish with column type Date (YMD). Then format as Date.
Numbers stored as text	creditLimit, amount, buyPrice, MSRP, priceEach	Check with =ISNUMBER(). If text: multiply by 1 in a helper column, or Text to Columns. Format as Number/Currency.
Missing shippedDate	orders — Cancelled / In Process / On Hold rows	This is legitimately blank (order never shipped). Leave blank; NEVER fill with 0 or today's date.
Duplicates check	every table's key column	COUNTIF check e.g. =COUNTIF(tblOrders[orderNumber],[@orderNumber])>1, or Conditional Formatting → Highlight Duplicates. (This dataset should have none — prove it.)
Extra spaces in text	customerName, contactFirstName	=TRIM(A2) in a helper column if needed; watch for names like 'Carine ' with a trailing space.

Checkpoint 1: every date column right-aligns and can be filtered by Year; no visible 'NULL' text anywhere; all 7 tables named correctly. Show your mentor before continuing.

5. Phase 3 — Build the Master Analysis Sheet (Days 4-6)

This is the heart of the project. You will enrich **OrderDetails** (2,996 rows) into a “flat” master table where every row has everything you need: product info, customer info, dates and money. Create a copy of the OrderDetails sheet, name it **Analysis**, and add these columns one by one.

5.1 Money columns (no lookups needed)

Revenue = quantity × price actually charged:

```
=[@quantityOrdered] * [@priceEach]
```

5.2 Pull product information (lookup into tblProducts)

```
=XLOOKUP([@productCode], tblProducts[productCode], tblProducts[productName])
=XLOOKUP([@productCode], tblProducts[productCode], tblProducts[productLine])
=XLOOKUP([@productCode], tblProducts[productCode], tblProducts[buyPrice])
```

Older Excel? Use VLOOKUP instead (column numbers count from productCode):

```
=VLOOKUP([@productCode], tblProducts, 2, FALSE) → productName
```

Now add **Cost** and **Profit**:

```
Cost =[@quantityOrdered] * [@buyPrice]
Profit =[@Revenue] - [@Cost]
```

5.3 Pull order information (lookup into tblOrders)

```
=XLOOKUP([@orderNumber], tblOrders[orderNumber], tblOrders[orderDate])
=XLOOKUP([@orderNumber], tblOrders[orderNumber], tblOrders[status])
=XLOOKUP([@orderNumber], tblOrders[orderNumber], tblOrders[customerNumber])
```

5.4 Date helper columns (from orderDate)

```
Year =YEAR([@orderDate])
Month =MONTH([@orderDate])
MonthName=TEXT([@orderDate], "mmm")
YearMonth=TEXT([@orderDate], "yyyy-mm")
```

5.5 Pull customer information (lookup into tblCustomers)

```
=XLOOKUP([@customerNumber], tblCustomers[customerNumber], tblCustomers[customerName])
=XLOOKUP([@customerNumber], tblCustomers[customerNumber], tblCustomers[country])
=XLOOKUP([@customerNumber], tblCustomers[customerNumber],
tblCustomers[salesRepEmployeeNumber])
```

5.6 Chain lookups to reach the sales rep and territory

```
SalesRep =XLOOKUP([@salesRepEmployeeNumber], tblEmployees[employeeNumber],
tblEmployees[firstName]) & " " & XLOOKUP([@salesRepEmployeeNumber],
tblEmployees[employeeNumber], tblEmployees[lastName])
OfficeCode=XLOOKUP([@salesRepEmployeeNumber], tblEmployees[employeeNumber],
tblEmployees[officeCode])
Territory =XLOOKUP([@officeCode], tblOffices[officeCode], tblOffices[territory])
```

Two lookups chained together (order detail → customer → employee → office) is exactly how analysts join 4 tables. If a lookup returns #N/A, check for stray spaces or text-vs-number mismatch in the key columns — wrap the key in VALUE() or TRIM() to diagnose.

Checkpoint 2: your Analysis table should now have roughly 20 columns and 2,996 rows, with no #N/A errors. Sum the Revenue column: it must equal **\$9,604,190.61**. If it doesn't, something is wrong — find it before moving on.

6. Phase 4 — Answer the Business Questions (Days 7-9)

Create a new sheet called **Pivots**. Build one Pivot Table per question (Insert → PivotTable, source = your Analysis table). Label each one with a bold title. These pivots become the engines behind your dashboard charts.

#	Business question	Pivot setup (Rows / Values)	Chart type
Q1	What are total revenue, profit and profit margin %?	No rows; Values: Sum of Revenue, Sum of Profit. Margin = Profit / Revenue (calculated field or formula next to pivot)	KPI cards
Q2	How is revenue trending month by month? Is the business growing?	Rows: YearMonth; Values: Sum of Revenue	Line
Q3	Which product lines make the most money — and which are the most profitable?	Rows: productLine; Values: Sum of Revenue, Sum of Profit; sort descending	Bar (sorted)
Q4	Who are our top 10 customers by revenue?	Rows: customerName; Values: Sum of Revenue; Value Filter → Top 10	Bar (sorted)
Q5	Which are the top 10 products by revenue?	Rows: productName; Values: Sum of Revenue; Top 10 filter	Bar (sorted)
Q6	Which countries generate the most revenue?	Rows: country; Values: Sum of Revenue; sort descending	Bar (sorted)
Q7	How do orders split by status? How many were cancelled?	Rows: status; Values: Count of orderNumber (distinct: use a helper pivot on tblOrders)	Doughnut
Q8	Which sales rep brings in the most revenue?	Rows: SalesRep; Values: Sum of Revenue; sort descending	Bar (sorted)
Q9	How does each territory (NA / EMEA / APAC / Japan) perform?	Rows: Territory; Values: Sum of Revenue	Column
Q10	Are customers paying what they owe? (Revenue billed vs payments received)	Compare SUM(tblPayments[amount]) with total revenue of non-cancelled orders; per-customer: pivot payments by customerNumber vs revenue by customerNumber	Table + conditional formatting

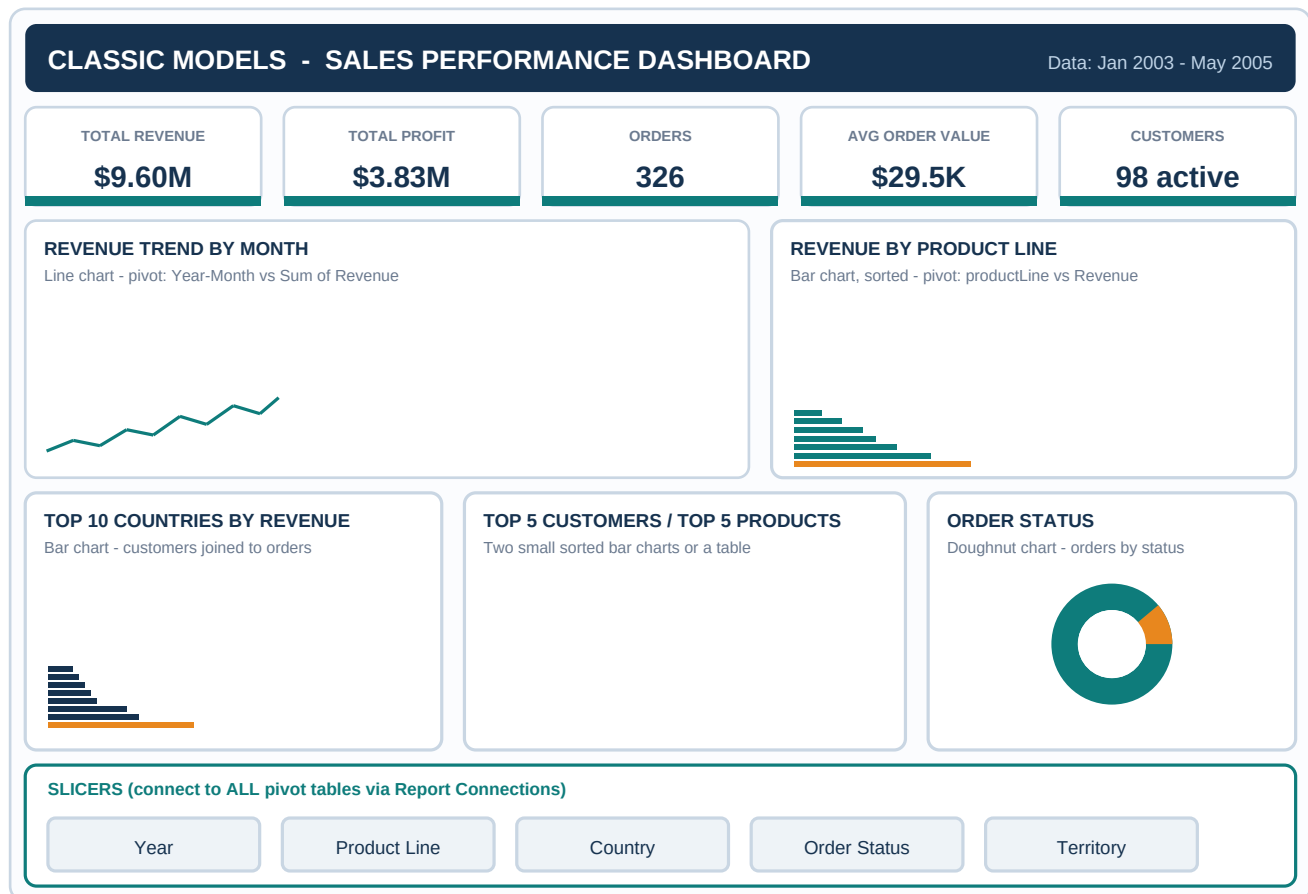
Extra formulas worth showing off (put these beside the pivots)

Average order value =SUM(Analysis[Revenue]) / COUNTA(tblOrders[orderNumber])
 Revenue in 2004 only =SUMIFS(Analysis[Revenue], Analysis[Year], 2004)
 Orders shipped late =COUNTIFS(tblOrders[shippedDate], ">0", ...) compare with requiredDate via helper column
 Avg days to ship =AVERAGE(helper column: [@shippedDate]-[@orderDate], only for shipped rows)

Checkpoint 3 — sanity numbers (your pivots must reproduce these): 2004 revenue ≈ **\$4.52M** (best year); Classic Cars is the #1 product line (≈ **\$3.85M**); top customer is **Euro+ Shopping Channel**; USA is the #1 country. If you get different winners, debug your lookups.

7. Phase 5 — Build the Dashboard (Days 10-12)

Create a sheet named **Dashboard**. Reproduce this exact layout — it is designed on a grid so everything aligns. Turn off gridlines on this sheet (View → untick Gridlines).



Build order

- 1. Title bar:** merge cells across the top, dark navy fill (#16324F), white bold 18pt text.
- 2. KPI cards:** five bordered cell blocks. Each shows a label and a big number linked to your pivots, e.g. `=GETPIVOTDATA("Revenue", Pivots!$A$3)` or simply `=Pivots!B4`. Format numbers as \$#,##0,,"M" for millions.
- 3. Charts:** create each chart from its pivot on the Pivots sheet (PivotChart), then cut & paste onto Dashboard. Remove chart junk: no gridlines, no legend where obvious, add data labels, sort bars descending.
- 4. Colour rule:** one accent colour (teal #0E7C7B) for normal series, orange (#E8871E) only to highlight the top item. Same fonts everywhere. This single rule makes dashboards look professional.
- 5. Slicers:** Insert → Slicer on Year, Product Line, Country, Order Status. Right-click each slicer → **Report Connections** → tick ALL your pivots so one click filters the whole page. Add a Timeline on orderDate.
- 6. Final polish:** align everything to the cell grid (hold Alt while dragging), equal gaps, test every slicer combination.

Test like a manager: click "2004 + Classic Cars + USA". Every card and chart should update instantly and tell a consistent story. If one chart doesn't move, its Report Connection is missing.

8. Phase 6 — Insights & Story (Days 13-14)

Numbers without words are not analysis. On an **Insights** sheet write 8-10 findings, each in the pattern *Observation* → *Number* → *So what?* Example: "Revenue peaked in Nov 2004 at \$1.06M (3× a normal month) — clear year-end seasonality; stock should be built up from September." Finish with 3 concrete recommendations for management.

9. Challenge Round (optional, for extra marks)

No instructions this time — you decide the approach. These separate good from great:

★	Challenge	Hint (only if stuck)
★	24 customers have never placed an order. Find them and their assigned sales reps.	COUNTIF of customerNumber in tblOrders = 0
★	Which products sell for far below MSRP? Compute average discount % per product.	1 - AVERAGE(priceEach)/MSRP per productCode
★★	Payments received (\$8.85M) are less than revenue billed. Which customers owe the most?	Per-customer revenue (non-cancelled) minus per-customer payments
★★	Compute each order's shipping delay and flag orders shipped after requiredDate.	shippedDate - orderDate; only 1 order in the data shipped late — prove it
★★	Which customers exceed 80% of their credit limit based on unpaid balance?	Balance / creditLimit with conditional formatting
★★★ ★	Build a mini 'RFM' view: for each customer, days since last order, number of orders, total revenue.	MAXIFS for last order date, COUNTIFS, SUMIFS
★★★ ★	The 2005 data stops in May. Forecast full-year 2005 revenue and justify your method.	Monthly run-rate, or compare Jan-May 2005 vs Jan-May 2004 growth

10. Grading Rubric (100 marks)

Component	Marks	What earns full marks
Data cleaning & setup	15	All 7 tables imported, named, NULLs handled, dates are real dates, fixes documented in ReadMe
Master Analysis sheet	20	All lookups correct, zero #N/A, Revenue total matches \$9,604,190.61, tidy column names
Pivot analysis (Q1-Q10)	20	All 10 questions answered with labelled pivots; checkpoint numbers reproduced
Dashboard design	20	Matches blueprint: 5 KPIs, 5-6 clean charts, consistent colours, aligned layout, no gridlines
Interactivity	10	Slicers + Timeline connected to every pivot; everything updates together
Insights & recommendations	10	8-10 specific findings with numbers; 3 actionable recommendations
Viva / walkthrough	5	Can explain any formula and defend any number in the workbook
Challenge round (bonus)	+10	Any 3 challenges solved correctly (capped at 100 overall or shown as distinction)

11. Submission Checklist

- Workbook named **ClassicModels_Analysis_YourName.xlsx** and opens without errors
- Sheets in order: ReadMe → Dashboard → Insights → Pivots → Analysis → 7 data sheets
- No 'NULL' text, no #N/A, no #REF! anywhere in the workbook
- Total revenue on dashboard = \$9,604,190.61 (before any filter)
- All slicers tested: clicking any combination updates every visual
- Charts have titles, data labels where useful, and sorted bars
- Insights sheet: 8-10 findings + 3 recommendations, each backed by a number
- ReadMe lists every cleaning step you performed
- File size sensible (< 10 MB) and saved in .xlsx format

12. Sample Viva Questions (mentor's ammunition)

- Why did you calculate revenue from order details instead of using payments.csv? When would the two differ?
- Your XLOOKUP returned #N/A for some rows during development. Walk me through how you debugged it.
- Why is a Pivot Table better than SUMIFS for Q3 — and when is SUMIFS better?
- 2005 revenue looks low. Is the business shrinking? Defend your answer.
- Should cancelled orders be included in revenue? What did you choose and why?
- What does the profit margin of ~40% assume? What costs are we ignoring?
- If a new month of data arrives as a CSV, list the exact steps to refresh your dashboard.
- Which single chart on your dashboard would you show the CEO if you had 30 seconds? Why?

13. Suggested Timeline

Days	Milestone
Day 1	Workbook set up, all files imported as named Tables
Days 2-3	Data cleaned; Checkpoint 1 approved by mentor
Days 4-6	Analysis master sheet complete; Checkpoint 2 revenue check passed
Days 7-9	All 10 pivots built; Checkpoint 3 sanity numbers reproduced
Days 10-12	Dashboard built, slicers connected, design polished
Days 13-14	Insights written, challenges attempted, viva preparation

Appendix — Mentor's Answer Key (do not share with students)

All values computed directly from the 7 provided CSVs. Revenue/profit figures include all order lines (including the 6 cancelled orders, worth ~1-2% of total); if a student excludes cancelled orders they should state so explicitly — accept either with justification.

KPI	Value
Total revenue (all order lines)	\$9,604,190.61
Total cost	\$5,778,310.36
Total profit / margin	\$3,825,880.25 (39.8%)
Orders / order lines	326 / 2,996
Customers (total / with orders / never ordered)	122 / 98 / 24
Products in catalogue / actually sold	110 / 109
Average order value	\$29,460.71
Total payments received	\$8,853,839.23
Order date range	6 Jan 2003 - 31 May 2005
Average days order → ship / late shipments	3.8 days / 1 order late

Breakdown	Answer (revenue)
By year	2003: \$3.32M 2004: \$4.52M 2005 (Jan-May): \$1.77M
By product line (sorted)	Classic Cars \$3.85M; Vintage Cars \$1.80M; Motorcycles \$1.12M; Trucks & Buses \$1.02M; Planes \$0.95M; Ships \$0.66M; Trains \$0.19M
Top 5 customers	Euro+ Shopping Channel \$820.7K; Mini Gifts Distributors \$591.8K; Australian Collectors \$180.6K; Muscle Machine \$177.9K; La Rochelle Gifts \$158.6K
Top 5 products	1992 Ferrari 360 Spider red \$276.8K; 2001 Ferrari Enzo \$190.8K; 1952 Alpine Renault 1300 \$190.0K; 2003 Harley-Davidson Eagle Drag Bike \$170.7K; 1968 Ford Mustang \$161.5K
Top 5 countries	USA \$3.27M; Spain \$1.10M; France \$1.01M; Australia \$0.56M; New Zealand \$0.48M
Order status counts	Shipped 303; Cancelled 6; In Process 6; Resolved 4; On Hold 4; Disputed 3

Teaching notes

- 2005 is a partial year (ends 31 May) — a favourite trap. Students who call 2005 “a bad year” without noticing this need the forecasting challenge.
- Euro+ Shopping Channel + Mini Gifts Distributors = ~15% of all revenue: great opening for a customer-concentration-risk discussion.
- payments < billed revenue opens the receivables conversation (Challenge ★★).
- The single late shipment (vs requiredDate) shows operations are fine — a good example of a 'boring but important' finding.
- Common failure points to watch: revenue computed from payments.csv; lookups keyed on customer name instead of customerNumber; cancelled orders silently included/excluded without a note; slicers not connected to all pivots.